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Motivation Related to Work: A Century of Progress

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Work motivation is a topic of crucial importance to the success of organizations and societies and the well-being of individuals. We organize the work motivation literature over the last century using a meta-framework that clusters theories, findings, and advances in the field according to their primary focus on (a) motives, traits, and motivation orientations (content); (b) features of the job, work role, and broader environment (context); or (c) the mechanisms and processes involved in choice and striving (process). Our integrative review reveals major achievements in the field, including more precise mapping of the psychological inputs and operations involved in motivation and broadened conceptions of the work environment. Cross-cutting trends over the last century include the primacy of goals, the importance of goal striving processes, and a more nuanced conceptualization of work motivation as a dynamic, goal-directed, resource allocation process that unfolds over the related variables of time, experience, and place. Across the field, advances in methodology and measurement have improved the match between theory and research. Ten promising directions for future research are described and field experiments are suggested as a useful means of bridging the research–practice gap.

Keywords: motivation, resource allocation, goals, self-regulation, intrinsic motivation

Motivation related to work remains one of the most enduring and compelling topics in industrial/organizational (I/O) psychology. As part of the larger field of motivational science, motivation related to work examines fundamental questions about the influence of nonability person attributes (e.g., motives, traits, goals), work ecologies, and the mechanisms and processes involved in purposive action. Work motivation affects the skills that individuals develop, the jobs and careers that individuals pursue, and the manner in which individuals allocate their resources (e.g., attention, effort, time, and human and social capital) to affect the direction, intensity, and persistence of activities during work. At the same time, work motivation is a topic of critical importance to public policymakers and organizations concerned with developing work environments, human resource policies, and management practices that promote vocational adjustment, individual well-being, and organizational success. As such, work motivation stands at the nexus of society, science, and organizational success.

The increasing importance of motivation over the last century is reflected in both the number and nature of motivation-related publications that have appeared in *Journal of Applied Psychology* (*JAP*). As shown in Figure 1, the number of these publications (as indexed by keyword count) has increased over the decades. However, a more thorough perusal of these articles indicates that motivation (although always considered important) did not truly emerge as a major research topic in *JAP* until the mid-20th century, with the introduction of theories by Atkinson (1958), Bandura (1977), Carver and Scheier (1981), Deci (1975), Fishbein and Ajzen (1975), Hackman and Oldham (1975), Locke (1966), Vroom (1964), and others. As these research streams have matured, motivation-related publications in *JAP* have focused on the role of individual differences (e.g., traits and motives), resource allocation and self-regulation processes, and the sociostructural context of work. As *JAP* celebrates its centennial, theory and research on work motivation stands at an important crossroad. Recent changes in the nature of work and the workforce have increased the need for new knowledge that can be used to cultivate and sustain work motivation. These trends, along with the availability and increasing use of new methodologies to track work experiences, assess motives, and evaluate motivational processing across levels and over time suggest that the field is well-positioned (and overdue) for yet another round of innovation and growth.

With these developments in mind, the purpose of this article is threefold: (1) to provide a meta-framework for organizing and understanding the work motivation literature as it has evolved over the last century, (2) to highlight major contributions and themes in different parts of the meta-framework (with an emphasis on works published in *JAP*), and (3) to identify major trends and promising future research directions.

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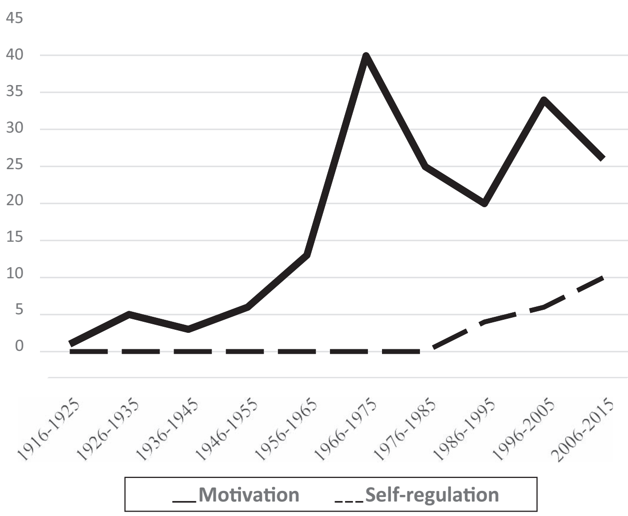

Figure 1. Number of *Journal of Applied Psychology* articles containing "motivation" and "self-regulation" keywords by time periods.

Toward a Meta-Framework: Conceptualizing the Phenomena and Rendering the Construct Space

Motivation is frequently used as an umbrella term meant to capture the dense network of concepts and their interrelations that underlie observable changes in the initiation, direction, intensity, and persistence of voluntary action. As in other complex areas of human functioning, researchers have pursued different aspects of the phenomena in ways that have sometimes led to confusion in understanding, studying, and managing work motivation. We propose a heuristic meta-framework (see Figure 2) that expands the often used process-content organization of relevant theories and motivation-related concepts into a tripartite organization: content, context, and process (Kanfer, Chen, & Pritchard, 2008). We base this organization on the observation that motivation theory and research in *JAP* and related journals has been driven by three fundamental questions: (1) What desires, wants, and needs elicit action? This is the classic question of what features of the individual spurs action, and pertains to the role of needs, universal motives (e.g., mastery, autonomy), and trait-based preferences. (2) What role do environmental factors play in motivation? Most theories of motivation recognize that behavior is also influenced by its context. Recently, Johns (2006) has suggested a multilevel perspective in which discrete contextual variables, such as job-related characteristics, sociostructural variables, physical conditions, and work events are nested within broader omnibus contextual variables, such as occupation, culture, and economic conditions. Most theory-driven research in *JAP* has focused on proximal or discrete contextual variables. (3) Through what psychological processes and mechanisms do person and environment factors affect the direction, intensity, and persistence of action? Process-oriented research focuses on the "how" of motivation. That is, what are the cognitive and affective mechanisms and processes by which individuals integrate person and environmental inputs to form goals, plans, and intentions (goal choice), and regulate their affect, attention, effort, and behavior for the purpose of goal accomplishment (goal striving). Several process-oriented theories currently popular in work motivation provide an integrative conceptualization of the processes involved in goal choice and goal striving.

Our organization of the field and major contributions follows from our systems approach, rather than adopting a strict historical perspective. As shown in Figure 2, within each section, theories, constructs, and research findings are clustered in terms of conceptual communalities and differentiated in terms of when we noted each approach to appear most prominently in the work motivation literature (often years after a theory's introduction in the broader psychology literature). In the content segment, we highlight theories and empirical findings on motives presumed fundamental for

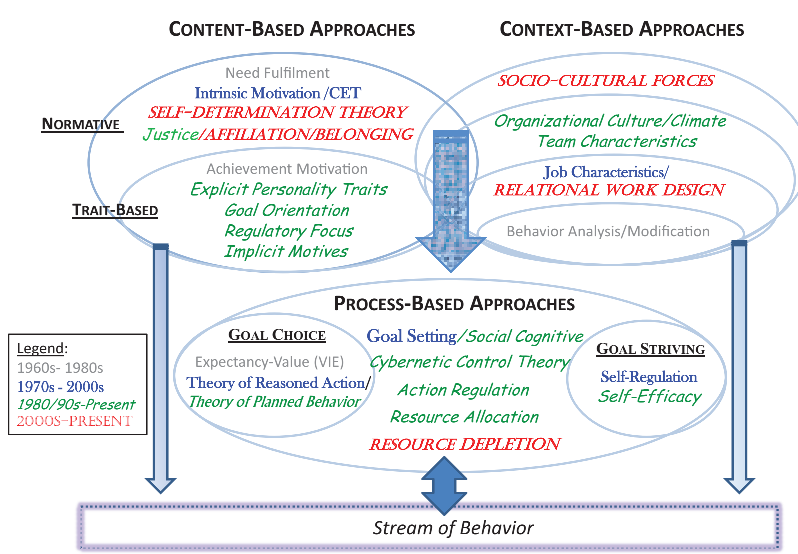

Figure 2. A heuristic meta-framework of work motivation construct networks and representative theories. See the online article for the color version of this figure.

all humans, followed by influential work on individual differences in personality traits and motivation orientations. In the context section, we use a multilevel organization to distinguish advances based on the level (e.g., job, team) at which contextual variables are typically studied. In the process section, theories and concepts are organized by their focus on goal choice, goal striving, and the dynamics of goal choice and striving over time. Figure 2 also illustrates how these areas are related to one another and to action. Although a number of studies provide evidence for the direct effects of content and context variables on work outcomes (as represented by the small arrows), the preponderance of evidence relating person and contextual variables to outcomes indicates that they exert their influence indirectly via their effects on motivational processes involved in goal choice and goal striving.

Progress in Work Motivation Research Over the Last Century: Content-Oriented Theories

Content-oriented theories specify the psychological traits, motives, tendencies and orientations that instigate motivational and volitional processes. These determinants reside within people (being either hardwired or learned) and give rise to enduring preferences for particular goals, strategies, and behaviors. Some person-based determinants are believed to represent universal sources of motivation for most people, whereas others exhibit more interindividual variance in their level and importance.

Universal Motives

Psychological discourse concerning universal motives can be traced back to works in the early 20th century that emphasized biological bases of motivation like instincts and drives (e.g., Freud, 1913; James, 1890). However, this older biological perspective fell out of favor because it was scientifically vague and had questionable utility. In its place, more comprehensive theories were introduced that included psychological and social needs, which precipitated an outpouring of motivation theories, as documented in the first work motivation text by Viteles (1953).

Need fulfillment. Needs are internal forces that are essential for supporting life and growth. Unmet needs create states of physical and psychological tension that energize action (Murray, 1938). Needs operate cyclically (i.e., they are not permanently satisfied), the strength of unmet needs corresponds to the tension they elicit, and satisfying unmet needs is posited to be rewarding. Although the needs described herein are common to everyone, there is between-person variance in motivation contingent on whether the needs are nourished or frustrated by the environment.

Maslow's (1943) hierarchical theory of needs posited behavior to be governed by five needs: physiological (e.g., nutrition), safety (e.g., stable employment), love (e.g., friendship), esteem (e.g., achievement), and self-actualization (e.g., self-fulfillment). Although all five needs fuel motivation, they are arranged hierarchically based on their prepotency (i.e., urgency for survival). Once physiological and safety needs are satisfied and hence become less salient, less prepotent needs (e.g., love and esteem) become influential. Despite the popularity of need theories during the 1960s and 1970s (e.g., Alderfer, 1969), empirical support was mixed at best (Wahba & Bridwell, 1976). Although there was some support for the structure and cross-cultural generalizability of Maslow's taxonomy (Ronen, 1994), there was little support for the prepotent nature of needs and the utility of need theories more generally (e.g., they do not predict specific behaviors; Campbell & Pritchard, 1976).

Some of these issues are redressed by contemporary need theories. For example, Self-Determination Theory (Deci & Ryan, 1985) specifies how satisfying basic needs for autonomy (i.e., personal causation), competence (i.e., mastery experience), and relatedness (i.e., genuine interpersonal connections) impact performance through intrinsic motivation.

Intrinsic motivation. Intrinsic motivation exists when people complete tasks because they are interesting and enjoyable (vs. solely as a means for gaining extrinsic incentives). Tasks that are interesting and enjoyable are believed to satisfy one or more of the universal needs for autonomy, competence, and relatedness. Although the concept of intrinsic motivation has been integral to theories of learning in educational psychology, an intriguing work-related proposition is that extrinsic incentives corrupt intrinsic motivation (Deci, 1975). Although some studies have indeed found that offering incentives can reduce intrinsic motivation (Pritchard, Campbell, & Campbell, 1977), especially when incentives are contingent (Pinder, 1976) and tasks are highly interesting (Daniel & Esser, 1980), other studies have failed to observe these undermining effects (Cameron & Pierce, 1994; Phillips & Lord, 1980). Further, although Deci, Koestner, and Ryan's (1999) meta-analysis found support for the undermining effects of incentives, Eisenberger, Pierce, and Cameron's (1999) meta-analysis found no clear-cut support for such effects in applied settings.

According to self-determination theory, a key driver of intrinsic motivation is whether or not task choice and behavior are perceived to be self-determined. Early findings testing cognitive evaluation theory (a central element of self-determination theory; Deci, 1975) found few detrimental effects for extrinsic rewards that were perceived to be informational (rather than controlling). Although self-determination theory is more aptly described as a grand perspective rather than a testable theory per se, parts of the theory have been successfully used to explain work motivation and behavior (Gagné & Deci, 2005). For example, Grant (2008a) found that firefighters and fundraisers worked longer and performed better when intrinsic motivation was high versus low (especially when paired with high prosocial motivation). In general, intrinsic motivation appears beneficial for performance on complex tasks, whereas extrinsic motivation aids performance on mundane tasks (McGraw & McCullers, 1979). Work environments that support (vs. frustrate) self-determination enhance other outcomes besides performance, such as creativity (Liu, Chen, & Yao, 2011), satisfaction (Deci, Connell, & Ryan, 1989), commitment (Greguras & Diefendorff, 2009), and retention (Liu, Zhang, Wang, & Lee, 2011). Meta-analytic results obtained by Cerasoli, Nicklin, and Ford (2014) further indicate that intrinsic motivation has a stronger impact on performance vis-à-vis extrinsic motivation when performance is defined in terms of quality (vs. quantity) and when extrinsic rewards are only indirectly tied to performance.

Justice motives. Employees also have a universal desire for fairness. Early work on equity theory (Adams, 1965) emphasized fair outcomes, finding that employees compare their inputs (e.g., effort) and outcomes (e.g., pay) with those of others, and any perceived imbalances create a psychological tension that they are motivated to reduce. As research in this area progressed, it was realized that the procedures used to determine outcomes, the respect conferred during interactions, and the candidness of explanations are crucial as well (Bies, 2001; Thibaut & Walker, 1975). Indeed, employee perceptions of fairness predict a wide variety of attitudinal, behavioral, and health outcomes (Colquitt, Conlon, Wesson, Porter, & Ng, 2001; Robbins, Ford, & Tetrick, 2012).

In addition to establishing different types of justice, researchers have explored why employees care about justice. In general, fairness is valued for instrumental reasons (e.g., it reduces uncertainty and fears of exploitation), relational reasons (e.g., it communicates positive social worth), and moral reasons (e.g., it aligns with normative standards of ethical conduct). It is therefore not surprising that fairness has indirect effects on behavior via instrumental mediators like perceptions of social exchange (e.g., Colquitt et al., 2013), relational mediators like social identity (e.g., Johnson & Lord, 2010), and affect-imbued morality mediators like anger (e.g., Weiss, Suckow, & Cropanzano, 1999).

Recent studies have explored ebbs and flows in daily justice experiences (Loi, Yang, & Diefendorff, 2009), the motives that drive justice behaviors (Scott, Colquitt, & Paddock, 2009), the characteristics of recipients that elicit fairness (Scott, Colquitt, & Zapata-Phelan, 2007), and the depleting and restorative effects of fair behaviors (Johnson, Lanaj, & Barnes, 2014). Initial evidence has also documented physiological and neurological processes that underlie justice experiences (Dulebohn et al., 2015; Yang, Bauer, Johnson, Groer, & Salomon, 2014). These findings suggest that justice motives are important contributors to work motivation.

Trait-Based Motives

As attention shifted from biological to psychological and social needs, researchers also began systematic study of interindividual differences in motives. Initial work mostly concerned achievement motivation, but trait-based approaches increased exponentially with the rise of the five factor model of personality in the 1980s. Since then, researchers have focused more attention on motivation traits and orientations that fall outside the five major trait categories.

Achievement, power, and affiliation motives. These motives have a long history in the motivation literature. Research on achievement motivation (i.e., the desire to accomplish something difficult by attaining high standards of excellence) began with Murray's (1938) pioneering work on needs. McClelland (1961)—a student of Murray's—popularized the idea of achievement motivation and spearheaded a large research program on this motive. In general, higher levels of achievement motive are associated with setting more challenging goals and attaining higher performance (Phillips & Gully, 1997). The other two motives (power and affiliation), which have received less attention, are more social in nature. The former motive refers to preferences for seeking and occupying positions of high social power, and the latter motive refers to preferences for establishing and sustaining positive relations with others. Studies have found, for example, differences in motive profiles among entrepreneurs and managers (McClelland, 1965; McClelland & Boyatzis, 1982) and positive relations of motives with starting a firm and entrepreneurial success (Rauch & Frese, 2007; Wainer & Rubin, 1969). Research on motives waned, however, during the 1970s owing to criticisms about personality research in general (Mischel, 1968) and in I/O psychology specifically (Mitchell, 1979).

Personality traits. The introduction of the five factor model revitalized research interest in delineating the effects of personality on motivation and performance. This interest was bolstered by Barrick and Mount's (1991) meta-analytic findings that showed conscientiousness and neuroticism to be reliable predictors of performance in most jobs, whereas other traits predicted performance on specific dimensions or in certain jobs (e.g., extraversion and sales). Observing reliable relationships of distal personality traits with performance led organizational researchers to more closely examine the motivational pathways by which such effects might occur. For example, Barrick, Stewart, and Piotrowski (2002) and Judge and Ilies (2002) showed that traits are linked to performance via the mediating effects of motivation (e.g., setting goals, gauging self-efficacy).

Other investigations have examined the influence of personality traits not included in the Big Five, such as action-state orientation (Kuhl, 1994; e.g., Diefendorff, Hall, Lord, & Strean, 2000) or have employed different integrative organizations of trait variables. Judge, Locke, and Durham (1997) integrated self-esteem, generalized self-efficacy, emotional stability, and locus of control to form the construct of core self-evaluation, or fundamental appraisals about one's worth and identity. Core self-evaluations predict a variety of affective, cognitive, and behavioral outcomes, including variables such as effort and persistence (Chang, Ferris, Johnson, Rosen, & Tan, 2012), that in turn mediate the effects of core self-evaluations on work outcomes (Ferris et al., 2013). Kanfer and Heggestad (1997) and Elliot and Thrash (2002) distinguish individual differences as markers of broader approach and avoidance motivations. Approach motivation guides behavior toward positive objects and outcomes, includes indicators like extraversion, and is associated with the behavioral activation system. Avoidance motivation, which guides behavior away from negative objects and outcomes, includes neuroticism is and associated with the behavioral inhibition system. Approach and avoidance motivation frameworks are useful because they have clearer ties to motivation than the five factor model, and have direct implications for goals that people set, the strategies they pursue, and the motivational skills they develop.

Motivation orientations. A related area of growing interest pertains to the role of inter-individual differences in the purpose of goal pursuits. For example, regulatory focus theory (Higgins, 1997) posits that goals may be pursued by maximizing gains to ensure success (promotion focus), or by minimizing losses to avoid failure (prevention focus). These foci are rival means for accomplishing the same ends, yet they have important implications for performance at work (Lanaj, Chang, & Johnson, 2012). For example, whereas prevention focus is a better predictor of safety (Wallace & Chen, 2006), promotion focus is a better predictor of productivity and also tends to foster greater well-being because maximizing gains elicits stronger positive emotions than minimizing losses (e.g., joy vs. relief; Ferris et al., 2013).

People also differ in the purpose of goal pursuit. Educational psychologists in the 1980s (e.g., Dweck, 1986; Nicholls, 1984) identified two reasons for goal pursuit: to develop competence (learning goal orientation) versus demonstrating competence so as to receive favorable appraisals from others (performance goal orientation). Learning goal orientation derives from incremental beliefs (i.e., abilities are malleable) and self-evaluations based on past performance progress, whereas performance goal orientation owes to entity beliefs (i.e., abilities are fixed) and evaluations relative to others. These beliefs constrain the goals that people set and the strategies they use (DeShon & Gillespie, 2005). Performance goal orientation is further divided into performance-prove (i.e., to prove one's competence for positive appraisals from others) and performance-avoid (i.e., to avoid appearing incompetent and evade negative apprasials; VandeWalle, 1997). It is typically found that learning goal orientation enhances performance, whereas performance-prove goal orientation has weak effects and performance-avoid goal orientation has detrimental effects (Payne, Youngcourt, & Beaubien, 2007).

Summary of Content-Based Motivation

Although midcentury research findings failed to support basic tenets of Maslow's need fulfillment theory, the intuitive appeal of the theory and the lack of alternative formulations contributed to its lingering popularity. Intrinsic motivation and justice theories have now filled the void. In contrast to Maslow, these theories focus exclusively on higher order needs (e.g., autonomy, justice), and the effects that thwarting or satisfying these motives have on work motivation.

Findings based on these theories are generally supportive, though the conditions that thwart motive satisfaction and work motivation do not appear uniform across all individuals. In a similar vein, the five factor model offers an appealing framework for identifying broad personality traits on which all people may be characterized. Research findings in this area provide support for the importance of individual differences in conscientiousness to work motivation and performance.

Recent theory and research on motivation orientations goes one step further to address how the way that an individual construes a work goal (the reasons for action) affects motivation and performance. In contrast to universal and personality trait approaches, motivation orientation is viewed as more variable, malleable and proximal to motivational processing. Research findings across different models indicate higher levels of motivation and performance among individuals who construe the reasons for action as appetitive (e.g., to learn, to attain) rather than preventive or protective. As Tett and Burnett (2003) suggested, it is important to understand how outcomes of these orientations that occur over time may shape the expression of motives and personality traits, and so contribute to popular conceptions of occupation-or culture-specific personalities and behaviors.

Context-Oriented Theories

Context-oriented theories focus on features of the environment that affect motivation and performance via their provision of affordances and constraints for motive satisfaction. Broad contextual variables (e.g., one's occupation) affect the nature of work and the environment in which work is performed. Work-specific contextual variables (e.g., supervisor support, coworker relations) reflect the proximal contextual determinants of work motivation and behavior. Context theories often highlight the strength of various features of the work environment (Meyer, Dalal, & Hermida, 2010). Contextual variables can influence motivational outcomes in two ways. Strong contexts and situations, in which individuals have little behavioral discretion, weaken the effects of individual differences by constraining variation in behavior. Contextual variables, such as team efficacy, can also moderate the influence of individual differences on motivational outcomes (Chen & Kanfer, 2006).

From Extrinsic Rewards to Internal Motivation

During the early 20th century, interest in identifying the motivating features of work was driven largely by the desire to increase efficiency and productivity in industrial production. Researchers working within the scientific management paradigm (Taylor, 1911) emphasized the use of extrinsic rewards (e.g., money) to enhance performance. Although extrinsic rewards can motivate higher performance (Rynes, Gerhart, & Minette, 2004), by the mid-20th century this narrow, mechanistic approach to motivation gave way to theories that emphasized the psychological pathways by which specific features of the job enhance motivation and performance. For example, Herzberg, Mausner, and Snyderman (1959) proposed a theory that organized job characteristics into two major groupings: hygienes (e.g., pay) and motivators (e.g., responsibility). Although empirical evidence does not provide strong support for this classification scheme (House & Wigdor, 1967), it was a pioneering attempt to identify those characteristics related to internal work motivation.

Task and Job Characteristics

In the mid-1970s, Hackman and Oldham (1975, 1976) introduced the job characteristics model. The impact of this theory was immediate and long-lasting, stimulating hundreds of studies and serving for over 2 decades as the scientific foundation for job redesign interventions to enhance work motivation. Building on Turner and Lawrence (1965), job characteristics theory specified five key features of a job (i.e., skill variety, task identity, task significance, autonomy, and feedback) and the psychological pathways by which these features operate (i.e., meaningfulness of work, experienced responsibility, and knowledge of results). Designing jobs that elicit these psychological states is posited to create a context in which "performance becomes its own reward, a virtuous self-perpetuating cycle of positive work motivation powered by self-generated rewards" (Hackman & Oldham, 1976, p. 256). The theory also acknowledged the moderating role of individual differences (i.e., growth need strength) on context–motivation relations.

Accumulated evidence for the validity of Job Characteristics Theory is largely supportive (Humphrey, Nahrgang, & Morgeson, 2007), though issues related to the measurement of job characteristics remains an important qualification of findings. Consistent with theories of self-determination and prosocial motivation, Humphrey et al. (2007) found task meaningfulness to be the most important mediator of job characteristics effects on work motivation. However, as Humphrey et al. (2007) and others have noted, the theory focuses on a limited number of contextual variables, and does not directly address the role of social and work-level variables, such as relational features of work roles (e.g., working in teams and with customers). Research into the effects of these variables led Humphrey et al. (2007) to extend the theory and examine accumulated evidence on the relationship between other motivational, social, and work condition variables on job attitudes and behavior. Their findings provide initial support for the role of additional contextual variables on job attitudes, and suggest the need for integrative research approaches that take into account theories of job stress and group/team influences on motivation and behavior.

Group-and Team-Level Influences

A historically separate line of inquiry into the role of contextual variables on work motivation focuses on the broader sociostructural context in which work is performed. Findings from the Hawthorne studies conducted during the early 1930s provide a vivid illustration for the role that social organization and group norms play in employee behavior. During the mid-20th century, social psychologists began to systematically study group characteristics and processes that lead to reduced team member motivation (e.g., social loafing). As the use of teams increased toward the end of the 20th century, organizational research on the effects of social context on motivation and performance moved into high gear. Although many studies have focused on the effects of team-level variables on team-level performance, evidence has also begun to accumulate on the cross-level and moderating effects of a range of organization-level (e.g., climate) and team-level variables (e.g., cohesion, shared team identity) on team member motivational states (e.g., affective commitment, personal initiative) and outcomes (e.g., performance, citizenship behavior, training transfer; Chen, Kirkman, Kanfer, Allen, & Rosen, 2007; Dietz, van Knippenberg, Hirst, & Restubog, 2015; see also Mathieu, Hollenbeck, van Knippenberg, & Ilgen, 2017). Nonetheless, more research is needed to investigate how team-level processes (e.g., planning and coordination) influence an individual's task goals, how resources are allocated across team and individual goals, and the unique demands of team membership on different forms of self-regulation.

Summary of Context-Based Motivation

Hackman and Oldham's (1976) Job Characteristics theory spawned a sizable research literature demonstrating how key features of the job can affect internal work motivation through their effects on experienced meaningfulness and felt responsibility for performance outcomes. Research in leadership also provides further evidence for the impact of supervisory relationships on motivation and performance (see Lord, Day, Zaccaro, Avolio, & Eagly, 2017). Research in these areas complement content-based approaches by showing work motivation is not just a matter of autonomy, competence, and fairness, but also about performing meaningful work, having responsibility, and experiencing supportive supervision Recent changes in the organization of work (e.g., from individual work to working in teams and multiteam systems) and growing globalization underscore the importance of these findings for understanding an individual's motivation in team and multiteam systems characterized by complex goal hierarchies and interdependencies. To date, research on individual motivation in teams has been relatively sparse and concentrated largely on the performance effects of team-level phenomena (e.g., social loafing). Additional research is needed to understand how social dynamics and emergent collective states, such as cohesion, trust, team identification, and entrainment affect the goals and resource allocation policies of individuals working in teams. Yet even these factors represent only part of the context in which individuals work. Research is sorely needed to understand the impact of sociocultural differences on employee strategies for goal accomplishment and the influence of occupational characteristics on psychological states that may facilitate or diminish work motivation over time.

Process-Oriented Theories

Over the last half century process-oriented theories have evolved to view motivation as comprised of two interdependent subsystems: a system governing goal selection and a system governing goal enactment (e.g., Heckhausen & Kuhl, 1985; Lewin, Dembo, Festinger, & Sears, 1944). The Rubicon model (Heckhausen, 1991) provides a vivid metaphorical description of the relationship and distinction between these subsystems. Before crossing the Rubicon river, which signified the border of Rome and point at which no further weapons were allowed, Caesar was reported to have rationally considered a number of military options and plans for achieving victory in his war against the Roman Senate. His decision to proceed with his army into the river cemented his commitment to move forward—the proverbial point of no return from deliberation (goal choice) to implementation (goal striving).

Early process-oriented theories focused on goal choice. Fueled by the midcentury rise of cognitive, information-processing psychology, work motivation theories described how beliefs and cognitions (e.g., expectancies and instrumentalities) combine to yield behavioral intentions or motivational force. By the 1970s, however, researchers grew increasingly critical of choice theories that were restricted to rational decision making processes before entering the Rubicon. New theories of self-regulation (Bandura, 1986; Kanfer, 1977), action control (Kuhl, 1984), and goal implementation (Gollwitzer, 1990) highlighted the importance of planning and the cognitive-affective processes involved in goal striving. As described subsequently, the last few decades witnessed the development of new paradigms for the simultaneous study of goal choice and goal striving.

Goal Choice. Goals are internal representations of desired states that direct attention, organize action, and sustain effort aimed at achieving those states. Goals vary across a range of attributes (e.g., specific vs. vague, performance vs. learning) and are arranged hierarchically, such that distal "ends" or "be" goals at the top of the hierarchy (e.g., earn a PhD) are linked to proximal "means" or "do" goals at lower levels (e.g., pass comprehensive exams). Establishing higher level goals constrains the goals that emerge at lower levels, and lack of progress on lower-level goals causes people to revise or even abandon higher level goals (Johnson, Chang, & Lord, 2006). During the mid-20th century, motivation studies in *JAP* frequently examined how person and environmental features influence the formation of proximal work goals and intentions.

Expectancy theory. Cognitive theories of goal choice posit a psychological calculus by which people rationally weigh the subjective benefits and costs of different options prior to selecting a goal or desired outcome that can be expected to maximize pleasure and minimize pain. A popular formulation is Vroom's (1964) Valence-Instrumentality-Expectancy (VIE) theory, which proposes that goal choice is determined by subjective evaluations of (a) the level of satisfaction expected from achieving work outcomes (valence), (b) the likelihood of achieving those outcomes by attaining a particular level of performance (instrumentality), and (c) the likelihood that one (with effort) will reach a certain level of performance (expectancy). These evaluations interact to determine the motivational force of specific goals or courses of action. Although empirical studies published in *JAP* and elsewhere during the 1950s through the early 1970s were generally supportive of expectancy models (e.g., Georgopoulos, Mahoney, & Jones, 1957; Mitchell & Nebeker, 1973), a number of conceptual and methodological problems were noted (e.g., decision making is not necessarily rational, within-person processes were tested using between-person methods, lack of support for the multiplicative nature of the theory's components; Heneman & Schwab, 1972; Van Eerde & Thierry, 1996). These criticisms paved the way for revised models and new formulations that emphasize the link between goal choice and goal striving.

Theory of planned behavior. Ajzen's (1991) theory of planned behavior extended expectancy theory by taking into account the interpersonal context in which goals and intentions are formed. Like expectancy models, this theory posits that the motivation to pursue a course of action is influenced by control beliefs (i.e., expectancy of being able to perform a behavior) and by attitudes (i.e., valence; Fishbein & Ajzen, 1975). The theory also includes subjective norms, which refers to perceived social pressure to pursue a course of action. According to the theory of planned behavior, behavioral intentions are jointly determined by perceived control, attitudes, and subjective norms.

The theory has been successfully applied to predict motivation to engage in a variety of work-related behaviors, such as job search (Wanberg, Glomb, Song, & Sorenson, 2005), volunteerism (Harrison, 1995), and managers' use of structured interviews (van der Zee, Bakker, & Bakker, 2002). A meta-analysis by Armitage and Conner (2001) was largely supportive of several tenets of the theory, yet also brought to light some key issues as well (e.g., problems with the conceptualization and operationalization of behavioral control, questionable incremental validity of subjective norm).

Goal Striving

Theories of goal choice set the stage for action, but they do not explain how individuals realize their goal (Kanfer, 1990). When goals are simple or well-learned (e.g., typing a letter), they can be accomplished quickly and without substantial effort. However, accomplishment of complex, protracted goals (e.g., obtaining a PhD, landing a new job) typically require people to modulate the direction and intensity of attentional effort, affect, and behavior over time and across component subgoals. These self-regulatory processes aid in maintaining goal intentions, evaluating goal progress, and making decisions about whether to persist, revise, or abandon the goal. Goal striving pertains to the activities that bridge the chasm between goals and performance. During the late 20th century, several major theoretical formulations of goal striving were proposed, including Bandura's (1986) social–cognitive theory, Carver and Scheier's (1981) cybernetic control theory, Frese and Zapf's (1994) action regulation theory, Heckhausen's (1991) Rubicon model, and Kanfer's (1977) self-regulation model.

Self-regulation. Theorizing by Bandura (1977), Carver and Scheier (1981), and Kanfer (1977) describe the psychological processes by which individuals strive for goal attainment. In these models, self-regulation involves three interrelated sets of activities: self-monitoring, self-evaluation, and self-reactions. *Self-monitoring* refers to the attention individuals give to events, behaviors, and feedback related to the goal. Failures in self-monitoring preclude the opportunity to evaluate goal progress. *Self-evaluation* refers to the comparative evaluation of the goal state to the current state, or goal progress. *Self-reactions* are the natural consequence of self-evaluations and encompass the affective and motivational responses toward discrepancies between desired and goal states. These self-reactions serve both an informational and motivational function in the self-regulation cycle.

Bandura's (1977) theory of self-efficacy further builds on and expands the self-regulation framework by delineating the role of self-efficacy in goal choice and goal striving. For example, self-efficacy moderates the impact of behavioral intentions on action.

Integrative Approaches

Self-regulatory formulations of goal striving altered the direction of work motivation research, as reflected in the publication of several hundred *JAP* articles over the last 35 years on different aspects of goal choice and goal striving. During this period, integrative formulations by Locke and Latham (1990), Kanfer and Ackerman (1989), Gollwitzer (1990), Vancouver (2008), and others began to focus specifically on the intersection of goal choice and goal striving.

Goal setting theory. From the early 1970s through the end of the century, Locke and Latham's (1990) goal setting theory was the leading theory of work motivation in I/O psychology. Goal setting theory emphasizes the link between goal attributes and action. Results from numerous lab studies and field experiments indicate that performance is highest when goals are specific (vs. "do your best"), difficult (vs. easy; Mento, Steel, & Karren, 1987), assigned using a "sell" (vs. "tell") approach (Latham, Erez, & Locke, 1988), and when they are coupled with performance feedback (Erez, 1977) and high goal commitment (Klein, Wesson, Hollenbeck, & Alge, 1999). Goals are posited to aid performance because they direct attention to goal-relevant activities, mobilize and sustain effort, and promote the use of task-relevant knowledge (Locke, Shaw, Saari, & Latham, 1981).

At work, goals are typically set by organizational representatives such as a supervisor; however, under most circumstances, employees also hold self-developed goals. As a result, the goals that direct actions are often an admixture of objectives held by the organization and by the employee. Goals that are set by the supervisor and "assigned" to employees are considered external goals, and the extent to which employees adopt the goal as their own reflects an internalization process (Frese & Zapf, 1994; Zacher & Frese, 2015). The redefinition of an assigned goal as one's own goal depends in part on task characteristics (such as clarity) and the extent to which there are conflicts between the individual's goals and the assigned goal.

Task complexity is an important boundary condition on the positive effects of difficult, specific goals, such that effects on performance are weaker when complexity is high versus low (Wood, Mento, & Locke, 1987). When tasks are complex and/or individuals have not developed the requisite skills to perform them, setting a difficult performance goal siphons critically needed attentional effort away from learning how to master the task. For example, Kanfer and Ackerman (1989) found that performance suffered when lower-ability individuals were assigned performance goals in the early stages of learning a complex air traffic simulation task, yet such goals had beneficial effects on performance once they became sufficiently skilled. When tasks are complex and/or individuals lack the required skills, it is better to assign a learning goal or a nonspecific "do your best" goal (Winters & Latham, 1996).

The earliest investigation of self-regulatory processes in the work domain occurred with the integration of goal setting theory and social–cognitive theory (Locke, Frederick, Lee, & Bobko, 1984). Since then, researchers have used the integrated goal setting/self-regulation framework to examine a variety of motivational phenomena, including the influence of self-efficacy on goal choice, the dynamics of self-regulation (e.g., Keith & Frese, 2005; Kozlowski & Bell, 2006), the impact of goal orientation on self-regulatory activities (e.g., DeShon & Gillespie, 2005; VandeWalle, Brown, Cron, & Slocum, 1999), the process by which individuals suppress distracting and goal-irrelevant information during goal striving (e.g., Johnson et al., 2006), and the self-regulatory strategies by which individuals accomplish multiple goals (e.g., Schmidt & DeShon, 2007; Sun & Frese, 2013).

Nonmechanistic versus mechanistic theories of motivation. Work on the self-efficacy–performance relationship during the last 15 years has rekindled a lively, long-standing debate on how to conceptualize motivation. For example, Hull (1943) argued that nonmechanistic concepts are based on an infinite regress of homunculi (homunculus in the psyche of a person presented new goals, new ideas, or new approaches, but then each homunculus has a psyche of its own and needs another one inside, leading to an infinite regress; cf. Verbruggen, McLaren, & Chambers, 2014). In contrast, nonmechanistic theorists (e.g., Tolman, 1949) argued against the primacy of stimulus–response concepts embodied by the use of technical analogues such as a switchboard or a thermostat, but rather for understanding action based on cognitive maps and the development of goal-oriented behaviors.

At the heart of this issue is the need to provide a perspective on motivation and action that systematically accounts for both routine and agency. Modern nonmechanistic theorists criticize mechanistic approaches as being unable to account for why individuals develop higher goals after they have mastered an action or skill. For example, Bandura's (1977) nonmechanistic, agentic theory of self-efficacy emphasizes the positive causal influence of self-efficacy on task performance across a variety of tasks and settings. There is strong empirical support for Bandura's position that individuals often increase their aspiration level once they have achieved a certain level of performance (Phillips, Hollenbeck, & Ilgen, 1996). Further, although there is broad evidence from studies using between-subjects designs to support the notion that very low levels of self-efficacy adversely affect the voluntary initiation of action, there is considerable controversy about the effects of high levels of self-efficacy on subsequent effort. Goal setting and social–cognitive formulations posit that high levels of self-efficacy exert feed-forward effects that bolster the adoption of higher goals and associated higher levels of goal-directed effort. In contrast, cybernetic control theory findings highlight the importance of a discrepancy between a desired set-point and a current state (e.g., if I expect to do well on the next exam, I do not need to exert as much effort in preparation as if I expect to do badly in the upcoming exam).

Findings by Vancouver and colleagues (e.g., Vancouver & Kendall, 2006; Vancouver et al., 2001) using within-subject designs show that high levels of self-efficacy are associated with subsequently lower levels of effort in a learning context. Vancouver, Weinhardt, and Schmidt (2010) have further argued that computational modeling is a good way to develop parsimonious explicit theories without any surplus meaning. From a nonmechanistic viewpoint, however, this relates to the general Turing problem of when a simulation is just a simulation or rather is built on true functional relationships (Miller, Galanter, & Pribram, 1960).

In short, we believe that agency and routine coexist (e.g., Vancouver, More, & Yoder, 2008) and that the designs used in these theories provide different insights into motivational processes. For example, Schmidt and DeShon (2010) helped to reconcile these perspectives by showing that self-efficacy was negatively related to performance when task ambiguity was high (as suggested by control theory) but positively related to performance when task ambiguity was low (consistent with social–cognitive theory). Such work advances our understanding of when and how each perspective may be most useful in the context of motivation related to work.

Action regulation theory. Action regulation theory provides an integration of goal choice and goal striving that highlights the importance of plans as a means by which individuals bridge the gap between having a goal and instigating activities to accomplish the goal. Plans are mental simulations of actions that determine how goals can be achieved and represent an important mediator in the goal setting–performance relation (Earley, Wojnaroski, & Prest, 1987). Plans are necessary for all actions, although the time between planning and action may vary from a few milliseconds to years, and the time frame covered by plans may similarly vary from a few minutes (planning to order a meal in a restaurant) to many years (a career plan). Planning aids performance when it prompts deeper thinking about intended actions, leads people to develop alternative plans ("Plans B and C"), and in unstable situations (e.g., in unpredictable economic situations; Frese, Krauss, et al., 2007).

Resource allocation theories. Resource allocation theories (Kanfer & Ackerman, 1989; Naylor, Pritchard, & Ilgen, 1980) view motivation as a process by which personal resources are allocated across an array of possible actions (e.g., goals) in response to task demands, affect, incentives, and feedback. Building on theories of human information processing and self-regulation, Kanfer and Ackerman (1989) proposed a resource allocation model that integrated cognitive abilities and motivational processes involved in goal choice and attentional resource allocation during complex skill acquisition. Findings from a series of studies showed the beneficial effects of such activities only when there were sufficient attentional resources available for self-regulation. Although DeShon, Brown, and Greenis (1996) found that self-regulation did not necessarily require attentional resources when performing a dual-task, concerns about the resource demands of self-regulation have continued to occupy attention.

To date, most integrative research has focused on the attainment of a single goal, yet in the real-world individuals frequently distribute their effort across multiple goals (e.g., make $250,000 in sales and improve customer satisfaction). Such situations require people to shift their attention across different tasks over time. Early research by Atkinson and Birch (1970) examined the conditions that prompted individuals to switch attention across two tasks, but did not address the conditions that affect task-switching across multiple tasks across a defined period of time. Recent work by Schmidt and colleagues (e.g.,Schmidt & DeShon, 2007) investigating the impact of incentives and goal progress on resource allocation across tasks and theorizing by Vancouver et al. (2010) and Steel and König (2006) are likely to further advance the field.

Resource depletion. The question of whether self-regulatory activities deplete cognitive resources over repeated use lies at the heart of ego depletion theory (Baumeister, Bratslavsky, Muraven, & Tice, 1998). This theory proposes that individuals have a finite amount of resources at their disposal for regulating behavior, which are drained by sustained acts of self-control (e.g., completing complex tasks, suppressing off-task distractions). In the workplace, for example, it has been found that vigilantly monitoring for potential problems (Lin & Johnson, 2015), suppressing and faking emotional displays (Trougakos, Beal, Cheng, Hideg, & Zweig, 2015), and acting consistent with procedural fairness rules (Johnson et al., 2014) are resource depleting activities. Once depleted, task, citizenship, and voice behaviors suffer (e.g., Lin & Johnson, 2015; Trougakos et al., 2015) while deviant and unethical behaviors increase (e.g., Christian & Ellis, 2011; Lin, Ma, & Johnson, 2016).

To avoid the negative consequences of prolonged self-regulation, researchers have explored ways that employees can counteract depletion and replenish resources. For example, positive social events (Bono, Glomb, Shen, Kim, & Koch, 2013), high autonomy (Sonnentag & Zijlstra, 2006), and respites during the workday (Trougakos, Beal, Green, & Weiss, 2008) aid in resource recovery.

Personal activities outside of the office can also replenish resources, such as nonjob mastery experiences (e.g., participating in sports and hobbies; Sonnentag, Binnewies, & Mojza, 2008), powering off work-related smartphones and computers (Lanaj et al., 2012), high quality sleep (Welsh, Ellis, Christian, & Mai, 2014), and vacations (Fritz & Sonnentag, 2006). Although ego depletion theory has proven useful for predicting breakdowns in self-regulation at work, some basic tenets of the theory have been recently criticized (e.g., Inzlicht & Schmeichel, 2012) and lab-based experiments of ego depletion in particular have been challenged (e.g., Carter, Kofler, Forster, & McCullough, 2015). Additional research on self-control processes and associated changes in regulatory resource availability is needed to explore alternative explanations for observed effects.

Summary of Process-Based Motivation

During the last 50 years, *JAP* has published a large number of process-oriented work motivation articles. Findings show that adoption of organizationally desired work goals is most likely to occur when employees perceive ownership of assigned goals, they believe that goal accomplishment is possible, and when goal achievement affords the receipt of intrinsic and/or extrinsic outcomes valued by employees. Many studies document the positive effects of explicit difficult and specific goals on effort and performance. Several studies also suggest that the effectiveness of such goals also depends on alignment with nonconscious goals, task demands/task complexity, and goal construal (e.g., learning vs. performance). Recent work has focused on the nexus of goal choice and goal striving and the dynamics of goal striving across single and multiple goals over time. The Rubicon metaphor highlights the function of goal planning as providing a bridge between goal choice and striving, and suggests a new arena for motivational research in planning and action control (Frese, Mumford, & Gibson, 2015).

The cumulative nature of theory and research in the process tradition has led to a more nuanced view of work motivation as a goal-directed, resource allocation process that changes over time as a consequence of the reciprocal interactions that occur between the person and the context in which action takes place (Kanfer, 2012; Sonnentag & Frese, 2012). In contrast to early expectancy theories that viewed effort as a "cold" or calculative process, current formulations take into account the "hot" or affective processes that also influence how attention and effort are allocated (Mitchell & Daniels, 2003). The evolving conception of motivation as a resource allocation process highlights the importance of a dynamic interactionist perspective for understanding the sources of variability in goal accomplishment. Whereas the valence of a goal had long been considered a function of static traits, goal orientation and regulatory focus studies have shown that relatively simple manipulations can change how employees construe goals, with significant downstream effects on planning and self-regulation. From a practical perspective, progress in this area has led to the development of management practices and feedback interventions that support sustainable employee regulation of attention and effort across work tasks over time.

Research Trends and Future Research Directions

As our review attests, progress in work motivation has occurred in fits and starts, with different areas in the field waxing and waning at different times. Nonetheless, we believe there has been progress. There is also a growing demand for understanding motivation in the context of new challenges related to modern work. In this section, we discuss research trends in how we study work motivation, and provide a sampling of future research directions.

Better Matching of Theory to Methodology

The last century has witnessed increased concern about the correspondence between theory and research methodologies (Van Eerde & Thierry, 1996). During the 1970s, reviewers of expectancy-value research pointed out the inappropriateness of testing a within-person theory with between-person designs (Mitchell, 1974). Although this criticism was directed at expectancy-value research, the point is valid for most theories that posit a change in motivation as a function of person—situation interactions that unfold over time. Certainly, between-person studies remain useful for determining the impact of a specific factor at a specific time and place. However, the adoption of a dynamic, interactionist perspective has led more researchers to examine motivation and performance trajectories using multilevel longitudinal studies that permit examination of changes and trajectories in motivational variables due to intraindividual processes, interindividual differences, and context. The disconnect between theory and methods has also lessened thanks to the development of measurement techniques that are better positioned to capture motivational phenomena that operate outside people's awareness and control.

Affect and motivation. There is broad agreement that motivation involves both cognition and affect. Yet most theories of work motivation continue to accord cognition primacy and relegate affect to a supporting role. Over the last few decades, however, progress in the study of emotions and affect has given rise to theories of action that highlight the role of different affective processes during self-regulation (e.g., Carver & Scheier, 1990; Kuhl, 2000; Seo, Barrett, & Bartunek, 2004; Weiss & Cropanzano, 1996) and to studies that examine the relationship between affect and work-related behaviors, such as job effort (Foo, Uy, & Baron, 2009), engagement (Bledow, Schmitt, Frese, & Kuhnel, 2011), and organizational citizenship behavior (Yang, Simon, Wang, & Zheng, 2016). Additional research on the role of affective processes in goal revision and goal pursuit offer exciting new possibilities for extending current integrative work motivation theories and developing new practices to manage motivation following affective events, such as downsizing.

Interplay of implicit and explicit motivation systems. Motivation researchers have long wrestled with the idea that some motives, traits, and orientations may exist and operate outside awareness. McClelland (1985), an early advocate of the implicit motive system, found that implicit motives explained unique variance in job performance incremental to explicit motives. Nevertheless, I/O psychologists have mostly focused on explicit motives. Recent advances in measurement make the exploration of implicit content and processes far more tractable than in the past (Uhlmann et al., 2012). For example, conditional reasoning tests capture implicit motives via responses to inductive reasoning problems (James, 1998) and have been successfully used to measure implicit aggression and achievement motives (e.g., Frost, Ko, & James, 2007). Popular, noninterpretation-based techniques, such as word fragment completion tasks, capture the accessibility of implicit content in memory (e.g., Johnson, Tolentino, Rodopman, & Cho, 2010).

Accumulated evidence over the last 2 decades using new measurement techniques supports the existence of a nonconscious motivation system related to preconscious attentional processes in sensory systems, learning, and performance. The initial wave of research in this area focused on evaluating the variance explained in work attitudes and behaviors by implicit versus explicit goals and motives (e.g., Johnson & Saboe, 2011; Stajkovic, Locke, & Blair, 2006). We expect the next wave will examine the ways in which implicit and explicit phenomena interface, converge, and change over time to affect work outcomes. For example, Bing, LeBreton, Davison, Migetz, and James (2007) found that employees with strong implicit and explicit achievement motives take on demanding tasks, whereas those with strong explicit but weak implicit achievement motives seek out tasks that they can deflect responsibility for failure. Future research is needed to further tease apart the intersection of implicit and explicit goals and motives, especially possible mediated relations, and to delineate how implicit content and processes change over time.

Temporal dynamics. Our review indicates a growing concern for understanding motivation over time and for the role that time-linked variables play in motivation and action (Kanfer, 2012). Findings from within-person studies show substantial variability in motivational variables over time.

Over the last 20 years, interest in time-linked effects on work motivation has developed in a variety of directions. Two areas of theoretical importance pertain to the effects that changing task demands have on resource availability over the course of practice (Kanfer & Ackerman, 1989) and understanding how individuals shift their allocation of resources across goals over time (Schmidt, Dolis, & Tolli, 2009). In the context of a purposive goal, we know that individuals invest different levels of effort toward goal accomplishment over time as a function of goal progress and deadline proximity (Wanberg, Zhu, & van Hooft, 2010). A different, but related theme pertains to what Roe (2014) referred to as the "temporal footprint of work (p. 63)." Roe suggested that motivation is likely to be cyclical and to vary in part as a function of the tasks and cycle time associated with different occupations. A third direction in understanding the temporal dimension in motivation is to consider the meaning of time and the way that subjective time perspectives (past, present, and future) connect individuals to their work. Recent studies on occupational future time perspective (e.g., Zacher & Frese, 2009) and concepts of future work self (Strauss, Griffin, & Parker, 2012) suggest that goal choice and striving depends not only on current conditions and motives, but on anticipatory forethought about one's future work situation and career. Investigations into the way that individuals construe goals and events through the lens of time, the role that work experiences and organizational culture plays in these processes, and the pathways by which they affect motivation and work behaviors represent interesting new directions for research.

Motivational resources. The conceptualization of motivation as a resource allocation process has gained traction over the last few decades as a useful means by which to map motivational dynamics, yet several questions remain. One fundamental issue pertains to what resources are. Kanfer and Ackerman (1989) defined resources as attentional effort in terms of cognitive resources of limited availability. However, recent work on self-control often uses a subjective assessment of the employees' phenomenological experience of depletion rather than tracking actual changes in resources, and findings are mixed regarding the physiological basis of resources (Molden et al., 2012). Yet other resource models define resources more broadly in terms of the individual's access to and possession of social-psychological and material assets that may be influenced by factors beyond the individual's control. A second issue is to delineate when depletion-based effects are likely to occur, because not all self-regulation activities appear to require resources. For example, Johnson et al. (2015) found that abiding by procedural justice rules was depleting, whereas interpersonal justice rules were not, suggesting that effortful self-regulatory resource demands may be substantially less for performance of culturally entrained behaviors. Resource-based theories of work motivation would benefit from further work concerning the nature of self-regulatory resources and the boundaries of depletion. A final issue is to rule out alternative explanations for reduced performance following activities requiring high levels of attention and self-control. For example, reductions in performance may owe to conscious shifts from self-constraint to cues for rewards when people expect to be recognized for their earlier efforts (Inzlicht & Schmeichel, 2012), to implicit beliefs about willpower being finite (Job, Dweck, & Walton, 2010), or to psychological licensing (Lin, Ma, & Johnson, 2016). These alternative explanations must be accounted for when examining depletion-based effects.

The self. One of the most striking developments in the *JAP* over the last century pertains to the increasing focus on the self. One line of research has examined the impact of individual differences in self-representations (personality traits) and implicit self-theories on goals and motivational processes and outcomes. Another productive stream of research looks at the effects of an individual's beliefs about their self-determination on work engagement and performance. A third line of inquiry derived from social–cognitive theorizing highlights people's ability to direct and control their behavior through self-regulatory processes.

Despite the plethora of studies investigating trait and state differences in self-related variables, researchers have yet to fully integrate the self in theories of work motivation. For example, theorizing and research on the autobiographical self (McAdams, 2013), possible future selves (Markus & Nurius, 1986), and other reflective self-construals (Markus & Kitayama, 1991) may be fruitfully applied to understanding how individuals construe work events (Fujita & Carnevale, 2012) and relationships. An important insight is that the self is socially constructed and multifaceted (Markus & Wurf, 1987). For example, employees can define themselves based on personal characteristics that distinguish them from others, based on ties to specific relational partners (e.g., a supervisor), or based on membership in salient groups (e.g., a work team or company; Brewer & Gardner, 1996). Accounting for these different personal, relational, and collective selves is essential because the motives and values that drive behavior shift depending on what aspect of self is currently salient. Future research is needed to identify the unique antecedents and consequences of employees' different self-definitions, and steps that organizations and managers can take to ensure that these self-definitions are appropriately aligned with task and company goals. The role of the self and identity in work motivation represents a promising new direction for future research.

Proactive engagement. The nature of motivation theories often reflect the impact of major changes in economics, technology, and organizational structures (e.g., contrast the difference between individual work processes in the early part of the 20th century and team-oriented work that has dominated the literature since the 1980s). Recent changes in the nature of work emphasize the need to be proactively engaged because many jobs (e.g., service jobs) cannot be formalized in detail and because accelerated technological changes imply that people have to preemptively prepare themselves for future changes, which has implications for job design and active changes of jobs in the sense of job crafting (Frese, Garst, & Fay, 2007; Wrzesniewski & Dutton, 2001) and active career development (Hall, 1996). This has created a need for research on the intersection of proactivity and engagement, which are interrelated (Hakanen Perhoniemi, & Toppinen-Tanner, 2008; Salanova & Schaufeli, 2008).

We define *proactive engagement* as a dynamic, self-directed syndrome of cognitive, affective, and motivational states characterized by high levels of vigor, dedication, and absorption that propel and sustain goal-directed activities (Grant & Parker, 2009; Parker, Williams, & Turner, 2006; Schaufeli & Bakker, 2010; Tornau & Frese, 2013). Investigation of proactive engagement requires identifying the job resources (e.g., autonomy, social support) needed for proactive engagement, and the consequences of such engagement for employee well-being (e.g., burnout, work-family conflict). A nascent body of work suggests an important link may exist between proactive work engagement and well-being during later adulthood (Schooler, Mulatu, & Oates, 2004). Investigation of proactive engagement also requires reconsidering the relations between affect and motivation (Bindl, Parker, Totterdell, & Hagger-Johnson, 2012: Sonnentag & Starzyk, 2015).

Although most theories of work motivation accord affect a service role in motivation, there is a growing literature using blended approaches to the study of motives (e.g., personal initiative) and affective states (e.g., flow). Given the importance of discretionary and proactive work behavior for personal and organizational success, there is value in further understanding the motivational processes underlying proactivity and personal initiative (Frese & Fay, 2001).

Motivation across the life span. Until recently, motivation theories had little to say about the effects of age-related changes on work motivation and behavior. However, growing age diversity in the workforce requires more attention to identifying age-related differences in motivational processing and key motivational levers at different points in the lifecycle (Kanfer & Ackerman, 2004). Research findings show complex age-related differences in work motives (Kooij, deLange, Jansen, Kanfer, & Dikkers, 2011), regulation of negative emotions (Scheibe, Sheppes, & Staudinger, 2015), and feedback orientation (Wang, Burlacu, Truxillo, James, & Yao, 2015). Additional research is needed to understand the nature of potential age-related differences in other motivational variables and processes, such as planning and self-regulatory strategies for work goal attainment.

Developmental perspectives also highlight the importance of understanding work motivation at different points across the individual's lifetime. The lion's share of theory and research over the last century has focused on motivation during work rather than the motivation to work (Kanfer, Beier, & Ackerman, 2013). Over a lifetime of employability that may last 50 years or more, individuals are likely to experience multiple periods of unemployment across their career. Although individuals are encouraged to take greater control over their careers, there is a need to better understand how communities and human resource management practices (e.g., recruitment, leadership) affect an individual's motivation to seek new employment, engage in professional development, and to continue working. Studies that focus on motivation to work at different points in the life course (e.g., school-to-work; following job loss and/or injury; nearing retirement) holds promise for informing public policymakers concerned with maintaining high levels of workforce employment in all age segments and for organizations that face increasing competition for a growing nontraditional workforce.

Contextual influences on motivation. Person-centric approaches to work motivation have substantially broadened the meaning of context to include the effects of culture, sociobehavioral norms associated with occupations, organizations, and work units, and nonwork demands on behavior and job performance. For example, findings by Mitchell, Holtom, Lee, Sablynski, and Erez (2001) on job embeddedness suggest that an individual's community involvement represents an important determinant of motivation for remaining at a job. Findings from the teams literature further suggests that team social dynamics play an important role in resource allocations. As the use of teams becomes ever more common and workforce diversity increases, multilevel research is needed to understand how the culture and relational context in which work is performed affects different aspects of work motivation, and the role of these factors in entraining distinct work motivation strategies over time.

Opportunities. Advances in basic science and changes in the way work is performed continue to create a wide-range of new opportunities and challenges. Along this line, we suggest two recent developments likely to have increasing influence on future work motivation research and practice. On the scientific front, findings in the neurosciences (cognitive and affective) suggest that motivation researchers will need to revisit the way we think about cognitive processes, affect, and choice during goal pursuit (Reeve & Lee, 2012). Embodied cognition research, for example, provides evidence for brain-cognition linkages between low-level physical states and higher order social–cognitive processes, and recent findings suggest that mind–body interactions influence goal activation and behavior during goal pursuit (Zhang & Risen, 2014). Such findings underscore the importance of sociality and all forms of context in work motivation. Over the long run, we expect advances in these areas will play a major role in the transformation of current cognitive models of work motivation (based largely on 20th century cognitive psychology). A second development with practical implications pertains to the integration of multiple technologies into the work experience, and the growing implementation of complex technological systems in diverse industry sectors, such as health care, homeland security, transportation management, and the military. Theory and research in human system integration provides a useful framework for understanding the multiple pathways by which person-technology interactions contribute to motivation and performance. Because many of the jobs in these systems are considered high-stakes, strategies to mitigate motivational deficits in these systems that are often associated with boredom, cognitive fatigue and job stress represent a critical challenge for the field.

Summary and Final Thoughts

Our review of work motivation theory and research suggest a maturing field in the early stage of major transformation. In contrast to earlier decades, I/O researchers currently think less about work motivation as an "on–off" phenomenon in which people respond in a constant manner to a motivational intervention, but rather as a dynamic, goal-directed, resource allocation process that unfolds over the related variables of time, experience, and place. Contemporary models of motivation as a dynamic resource allocation process over time underscore the importance of studying motivation over meaningful cycles of time and in connection with events that have affective significance. These models are well-suited for integration with advances in affect and neuroscience for investigating the reciprocal relations between personal attributes, experiences, and the environment and for their relations to resource allocation patterns across activities over time, with potential negative (resource depletion) and positive (resource accumulation) consequences to the self and performance. Three features distinguish this contemporary perspective from older views, such as the (1) primacy of goals, (2) emphasis on goal pursuit and associated affective processes, and (3) conception of motivation as an active process in which people take personal initiative, exert voice, and take charge of their motivation.

Findings in *JAP* and elsewhere document the critical role that various aspects of goal choice and goal striving play in motivation and the myriad ways in which person and contextual factors influence motivational processing. Goals form the nexus through which the "why" of action (variously defined as needs, motives, desires, or interests) connect with the "how" of purposive action. Some theories seek to explain and predict why different goals are adopted and the forces that bind an individual to a particular direction of action. Other theories focus on how goals and intentions are formed and modified. Newer integrative formulations seek to bridge the why and the how and to account for the active role of the self, plans and strategies, affect, and both the explicit and implicit content and processes that underlie work motivation.

Although this review has focused primarily on progress in work motivation theory, we would be remiss if we did not acknowledge the importance of research that demonstrates how scientific knowledge can be used to benefit individuals, organizations, and society. Field experiments that document the impact of various theory-based interventions, such as goal setting (Latham & Yukl, 1975), work redesign (Grant, 2008b), expectancy training (Eden & Shani, 1982), self-management training (Frayne & Geringer, 2000), and goal orientation training (van Hooft & Noordzij, 2009) can be highly influential in the uptake of new approaches in organizations, We encourage more of these studies as a means of reaffirming the value of motivation theory in the psychology of work and organizations and building a stronger foundation for constructive transformation of the field.

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